

DCF Valuation Professional v3.0.5

AICA Level 2 Batch 104 Capstone

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AI-assisted offline and online DCF valuation utility

Problem Statement

DCF models are often manual and inconsistent

Financial extraction from Schedule III statements is difficult

Unlisted valuation requires bottom-up beta and professional assumptions

Need for privacy-aware AI-assisted automation

Project Objective

Build reusable desktop utility

Support listed and unlisted companies

Import Excel/PDF financials

Generate DCF assumptions using offline rules or AI

Export complete valuation pack

System Architecture

Company setup and historical input

Financial statement import and review

Assumption generation engine

Formula-driven DCF engine

Excel, Word, PDF and PowerPoint outputs

AI Methodology

Offline deterministic assumption engine

Local AI through Ollama

Optional online AI through OpenAI/Gemini

AI assists; human review remains mandatory

Financial Methodology

$FCFF = EBIT(1-T) + D\&A - Capex - \Delta NWC$

WACC based on CAPM and cost of debt

Terminal value using Gordon Growth

EV to equity value bridge

Unlisted Company Module

Comparable company beta input

Unlever peer betas

Average unlevered beta

Relever to subject company D/E

Optional company-specific risk premium

Schedule III Import Controls

Financial-year column detection

P&L / Cash Flow / Balance Sheet context scoring

High / Medium / Low / Reject confidence

Review screen before posting figures

Privacy and Governance

Offline-first design

Online AI only after explicit consent

API keys not saved in project file

Machine-bound signed reuse key licensing

Outputs

Excel DCF model

Word and PDF valuation report

PowerPoint summary

Project JSON audit trail

Sensitivity and scenario analysis

Testing and Iteration

Windows testing

Ollama installation and model detection

Nuitka EXE compilation

Importer and timeout corrections

Final v3.0.5 release

Conclusion

AI improves speed, consistency and documentation

Valuation remains formula-driven and auditable

Designed for professional review, not blind automation

Suitable as AICA Level 2 finance automation capstone